



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
End Semester Examination I
Academic Year: 2024-25 (Semester – II)

Class: FY M.Tech.	Subject: Web and Network Technologies	Subject Code: PCC-PL-511
Date: 22.03.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/sharing of stationery, calculator, etc. is not allowed.
- Figures to the right indicate full marks.
- The symbols carry the usual meaning.

Q.N.		Marks	CO	BT
Q.1	a Discuss the three parts of an IP address.	4	CO 1-2	BT 1-2
	b Why is IP called connectionless? & unreliable - 6M	4	CO 1-2	BT 1-2
	c Discuss the process of communication using TCP/IP. 4M.	6	CO 1-2	BT 1-2
Q.2	a Create a website for the Department of Mechanical Engineering consisting of three to four webpages. Demonstrate the use of style file and javascript file in your html webpages. Demonstrate your results.	6	CO 2-3	BT 1-4

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Select proper web and internet network.
CO 2	Gain knowledge of Networking Protocols, Internet working concepts, History and Architecture.
CO 3	Understand working of WWW, HTTP, TELENET, JavaScript, and Ajax, PHP/MySQL, ASP.Net.
CO 4	Gain knowledge of Java Web Technologies.
CO 5	Select proper Web Security and Network Security.
CO 6	Understand the concept of XML and Cloud Computing.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
End Semester Examination (Semester - II)
Academic Year: 2024-25

Class: FY-M.Tech. (Common to All)

Subject: Project Management

Subject Code: AUD-901

Date: 30/04/2025

Time: 10:30 am to 12:30 noon

Max. Marks: 60

Note:

- Non-programmable calculator is permitted.
- Figures to right indicate full marks.
- This paper contains three pages

iii. This paper contains three pages

Q.No.	Question	Marks	CO	BT																																								
Q1	a State and explain Risk management planning.	05	CO1	BT2																																								
	b Differentiate between CPM and PERT.	05	CO2	BT1																																								
Q2	a Define: Float. Explain the measure of certainty in PERT.	03	CO2	BT3																																								
	Find out the time, variance and standard deviation of the project with the following time estimates in weeks:																																											
	<table><tr><th>Activity</th><th>Optimistic time estimate (to)</th><th>Most likely time estimate (tm)</th><th>Pessimistic time estimate (tp)</th></tr><tr><td>1-2</td><td>3</td><td>6</td><td>9</td></tr><tr><td>1-6</td><td>2</td><td>5</td><td>8</td></tr><tr><td>2-3</td><td>6</td><td>12</td><td>18</td></tr><tr><td>2-4</td><td>4</td><td>5</td><td>6</td></tr><tr><td>3-5</td><td>8</td><td>11</td><td>14</td></tr><tr><td>4-5</td><td>3</td><td>7</td><td>11</td></tr><tr><td>6-7</td><td>3</td><td>9</td><td>15</td></tr><tr><td>5-8</td><td>2</td><td>4</td><td>6</td></tr><tr><td>7-8</td><td>8</td><td>16</td><td>18</td></tr></table>	Activity	Optimistic time estimate (to)	Most likely time estimate (tm)	Pessimistic time estimate (tp)	1-2	3	6	9	1-6	2	5	8	2-3	6	12	18	2-4	4	5	6	3-5	8	11	14	4-5	3	7	11	6-7	3	9	15	5-8	2	4	6	7-8	8	16	18	10	CO2	BT3
Activity	Optimistic time estimate (to)	Most likely time estimate (tm)	Pessimistic time estimate (tp)																																									
1-2	3	6	9																																									
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2-4	4	5	6																																									
3-5	8	11	14																																									
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6-7	3	9	15																																									
5-8	2	4	6																																									
7-8	8	16	18																																									
	Find the probability that the project will be completed in 37 weeks or less. (use normal distribution table)																																											
Q3	a Explain the concept of crashing of a project.	03	CO3	BT2																																								
	The data for the project network and manpower requirement for a project is as follows. Obtain resource usage profile based on early estimate.																																											
	<table><tr><th>Activity</th><th>P</th><th>Q</th><th>R</th><th>S</th><th>T</th><th>U</th><th>V</th><th>W</th></tr><tr><td>Immediate Predecessor</td><td>-</td><td>P</td><td>P</td><td>Q</td><td>Q</td><td>R</td><td>S</td><td>T, U</td></tr><tr><td>Duration (Weeks)</td><td>3</td><td>4</td><td>6</td><td>6</td><td>4</td><td>4</td><td>6</td><td>8</td></tr><tr><td>Resource (M)</td><td>2</td><td>3</td><td>5</td><td>4</td><td>1</td><td>2</td><td>3</td><td>4</td></tr></table>	Activity	P	Q	R	S	T	U	V	W	Immediate Predecessor	-	P	P	Q	Q	R	S	T, U	Duration (Weeks)	3	4	6	6	4	4	6	8	Resource (M)	2	3	5	4	1	2	3	4	12	CO3	BT4				
Activity	P	Q	R	S	T	U	V	W																																				
Immediate Predecessor	-	P	P	Q	Q	R	S	T, U																																				
Duration (Weeks)	3	4	6	6	4	4	6	8																																				
Resource (M)	2	3	5	4	1	2	3	4																																				
	If at a time only 8 persons are available on each day, then level the resources accordingly. (note: use AON approach for network & Graph paper for usage Profile)																																											



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
End Semester Exam
Academic Year: 2024-25 (Semester -II)

Class: M.Tech.	Subject: Machine Learning	Subject Code: PEC-PL-518
Date: 05.05.2025	Time: 10:30 to 12:30	Max. Marks: 60

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. This paper contains two pages.

Q.N.		Marks	CO	BT																								
Q1	Explain the concept of reinforcement learning and its significance in machine learning. Include details on its fundamental principles, key components of a reinforcement learning system, and provide a real-world example illustrating its application.	6	CO 1	BT 1-3																								
Q2	Explain the concept of quartiles in detail and identify any outliers in the following data set: 8, 17, 23, 21, 30, 75, 12, 45, 38, 33	6	CO 1-3	BT 1-4																								
Q3	Explain the key steps of the Machine Learning Development Life Cycle (MLDLC) and discuss the significance of each stage in developing an efficient machine learning model.	6	CO 1-3	BT 1-3																								
Q4	For each column in the dataset, suggest an appropriate feature engineering technique that can be used to preprocess the column for machine learning. Justify your choice. <table><tr><th>S.No</th><th>Age</th><th>Education</th><th>Favourite colour</th></tr><tr><td>1</td><td>25</td><td>High School</td><td>Blue</td></tr><tr><td>2</td><td>32</td><td>Master's</td><td>Green</td></tr><tr><td>3</td><td>28</td><td>Bachelor's</td><td>Red</td></tr><tr><td>4</td><td>45</td><td>Ph.D</td><td>Yellow</td></tr><tr><td>5</td><td>22</td><td>High School</td><td>Blue</td></tr></table>	S.No	Age	Education	Favourite colour	1	25	High School	Blue	2	32	Master's	Green	3	28	Bachelor's	Red	4	45	Ph.D	Yellow	5	22	High School	Blue	6	CO 1-3	BT 1-5
S.No	Age	Education	Favourite colour																									
1	25	High School	Blue																									
2	32	Master's	Green																									
3	28	Bachelor's	Red																									
4	45	Ph.D	Yellow																									
5	22	High School	Blue																									
Q5	What is Complete Case Analysis (CCA) in the context of handling missing data? Discuss the key assumptions behind CCA, along with its main advantages and disadvantages.	6	CO 1-3	BT 1-3																								
Q6	Define univariate and multivariate imputation techniques used for handling missing data. Then, using the K-Nearest Neighbors (KNN) imputation method with $K = 2$, fill in the missing values in the dataset given below. <table><tr><th>Sr. No.</th><th>Feature1</th><th>Feature2</th><th>Feature3</th></tr><tr><td>1</td><td>12</td><td>- 24</td><td>15</td></tr><tr><td>2</td><td>- 20.5</td><td>20</td><td>25</td></tr><tr><td>3</td><td>23</td><td>18</td><td>- 20</td></tr><tr><td>4</td><td>- 18</td><td>30</td><td>24</td></tr></table>	Sr. No.	Feature1	Feature2	Feature3	1	12	- 24	15	2	- 20.5	20	25	3	23	18	- 20	4	- 18	30	24	10	CO 1-3	BT 1-4				
Sr. No.	Feature1	Feature2	Feature3																									
1	12	- 24	15																									
2	- 20.5	20	25																									
3	23	18	- 20																									
4	- 18	30	24																									

Q4	a	State and explain Appraisal Techniques.	06	CO																														
	b	The following information belongs to four investment opportunities: <table border="1"> <thead> <tr> <th></th><th>1</th><th>2</th><th>3</th><th>4</th></tr> </thead> <tbody> <tr> <td>Funds needed</td><td>(960,000)</td><td>(720,000)</td><td>(540,000)</td><td>(900,000)</td></tr> <tr> <td>Present value of cash inflows</td><td>1,134,540</td><td>866,800</td><td>672,280</td><td>1,045,940</td></tr> <tr> <td>Net present value</td><td>\$174,540</td><td>\$146,800</td><td>\$132,280</td><td>\$145,940</td></tr> <tr> <td>Project life</td><td>6 years</td><td>12 years</td><td>6 years</td><td>3 years</td></tr> <tr> <td>Internal rate of return (IRR)</td><td>16%</td><td>14%</td><td>18%</td><td>19%</td></tr> </tbody> </table> Due to limited funds (i.e, 20,00,000/-), all projects cannot be accepted. 1. Compute profitability index (present value index) for all the projects. 2. Rank the four investment projects according to (a). net present value (b). profitability index (c). internal rate of return 3. Which of the above investments should be undertaken? Assume that projects are indivisible and there is no alternative use of money allocated for capital budgeting.		1	2	3	4	Funds needed	(960,000)	(720,000)	(540,000)	(900,000)	Present value of cash inflows	1,134,540	866,800	672,280	1,045,940	Net present value	\$174,540	\$146,800	\$132,280	\$145,940	Project life	6 years	12 years	6 years	3 years	Internal rate of return (IRR)	16%	14%	18%	19%	09	CO
	1	2	3	4																														
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Internal rate of return (IRR)	16%	14%	18%	19%																														
Q5		Which are the different soft tools for Project Management. Explain in detail	07	CO																														

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO	On successful completion of this course, students should be able to:
CO 1	Understand the concepts and functions of project management.

NORMAL DISTRIBUTION TABLE

Area Under Standard Normal Distribution

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817

Q7	What are Z-score, Interquartile Range (IQR), and Percentile-based methods in machine learning? Explain how each method is used for outlier detection.	6	CO 1-3	BT 1-3
Q8	What do you understand by the term "curse of dimensionality"? Illustrate the concept with appropriate examples to explain its impact on data analysis and machine learning.	4	CO 1-3	BT 1-3
Q9	Describe regression-based and cluster-based modeling techniques used in machine learning. List and briefly explain the different types of models under each category.	10	CO 1-3	BT 1-4

== ALL THE BEST ==



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
ESE-Semester Examination II
Academic Year: 2024-25 (Semester – II)

Class: FY / M.Tech.	Subject: Advanced Concepts (PLM)	Subject Code: PCC-PL510
Date: 02.05.2025	Time: 10:30 to 12:30	Max. Marks: 60

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/sharing of stationery, calculator, etc. is not allowed.
- iv. Figures to the right indicate full marks.
- v. The symbols carry the usual meaning.

Q.N.

		Marks	CO	BT
Q.1	What is Action Handler and Rule Handler? What is quorum? Explain the following the handler. a) EPM-Create-relation b) EPM-Set-Property c) EPM-Check-object-properties d) EPM-assert-target-checked-in e) EPM-check-status	20	CO3	BT2, BT3
Q.2	a 1) What is rule base protection and object-based protection in access manager? 2) Write down the rule syntax? 3) What is access control list? 4) Explain how rule tree evaluate for effective acl by taking example.	10	CO3	BT1, BT2
	b 1) Explain Id Generation Rule (IPN) with example. 2) Explain naming rule and revision naming rule with example.	10	CO3	BT1
Q.3	a What is utilities in Teamcenter? Write down the following utilities with proper syntax and explain in brief. 1) clearlocks utilities. 2) generate meta data cache. 3) global transfer utilities.	08	CO1	BT1, BT2
	b What is organisation in PLM. Differentiate between person and user.	04	CO4	BT1

Q.4

What is change manager? Explain following term in change manager.

08

CO3

BT1,
BT2

- 1) Problem report (PR).
- 2) Engineering change request (ECR).
- 3) Engineering change notice (ECN).

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Exhibits the use of various elements of PLM.
CO 2	Selects proper architecture of PLM tools for an Industry.
CO 3	Administer for installing, configuring, integrating and efficient management of PLM tools.
CO 4	Select proper methodology for data transfer and implementation in CAD and PLM systems.

 Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Mechanical Engineering End Semester Examination Academic Year: 2024-25 (Semester – II)		
Class: FY M.Tech.	Subject: Web and Network Technologies	Subject Code: PCC-PL-511
Date: 06.05.2025	Time: 10:30 to 12:30	Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/sharing of stationery, calculator, etc. is not allowed.
- Figures to the right indicate full marks.
- The symbols carry the usual meaning.

Q.N.		Marks	CO	BT
Q.1	a	3	CO 1-2	BT 1-2
	b	4	CO 1-2	BT 1-2
	c	3	CO 1-2	BT 1-2
Q.2	a	5	CO 2-3	BT 1-2
	b	5	CO 1-2	BT 1-2
Q.3	a	6	CO 1-2	BT 1-2
	b	4	CO 1-2	BT 1-2
Q.4	a	5	CO 1-2	BT 1-2
	b	5	CO 1-2	BT 1-2
Q.5	a	20	CO 3-6	BT 1-5

— ALL THE BEST —

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Select proper web and internet network.
CO 2	Gain knowledge of Networking Protocols, Internet working concepts, History and Architecture.
CO 3	Understand working of WWW, HTTP, TELENET, JavaScript, and Ajax, PHP/MySQL, ASP.Net.
CO 4	Gain knowledge of Java Web Technologies.
CO 5	Select proper Web Security and Network Security.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
END SEM EXAMINATION
Academic Year: 2024-25 (Semester - II)

Class: First Year M.Tech (PLM)	Subject: Internet of Things (IoT) Elective- IV	Subject Code: PEC-PL-519
Date: 08.05.2025	Time: 10:30 - 12:30	Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/sharing of stationery, calculator, etc. is not allowed.
- Figures to the right indicate full marks.
- Indicate the Question Numbers while answering each question correctly.

Q.N.	Solve any TWO out of THREE sub-questions in Each.	Marks	CO	BT
Q.1	(a) Explain the concept of IoT, its characteristics, and how it enables smart applications like smart cities and healthcare systems.	06	CO1	BT5
	(b) Describe the architecture of IoT in detail, highlighting the roles of the perception, network, processing, application, and business layers.	06	CO1	BT4
	(c) Discuss the various communication protocols used in IoT, focusing on their characteristics, applications, and challenges.	06	CO1	BT6
Q.2	(a) Explain the architecture and working of Wireless Sensor Networks (WSN). Discuss their applications in modern industries.	06	CO2	BT2
	(b) What is Machine-to-Machine (M2M) communication? Explain its working, key components, and role in IoT with examples.	06	CO2	BT1
	(c) Define Interoperability in IoT. What are the challenges in achieving interoperability, and how can they be addressed?	06	CO2	BT1
Q.3	(a) Write a python program to calculate 'Area of Circle' with radius 5cm.	06	CO3	BT5
	(b) Draw the structure of Raspberry Pi and explain the advantages and limitations of Raspberry Pi compared to a traditional PC.	06	CO3	BT2
	(c) Draw and explain the SDN architecture. Also, compare SDN with traditional networking by highlighting 04 key differences.	06	CO3	BT2
Q.4	(a) What are the different types of cloud computing services? Explain each type in detail, including their structure and components.	06	CO4	BT1
	(b) What is Fog Computing? Discuss its various types and describe the scenarios in which fog computing is typically employed.	06	CO4	BT1
	(c) Explain the concept of 'Smart City' using IoT application through the cloud?	06	CO4	BT2
Q.5	(a) Define the Industrial Internet of Things (IIoT) and explain how it differs from traditional IoT.	06	CO5	BT1
	(b) What are the cybersecurity challenges associated with IIoT?	06	CO5	BT1
	(c) List and explain the benefits of integrating IIoT with PLM.	06	CO5	BT4

Course Outcomes (CO)

- CO1. To introduce the concept of Internet of Things and its horizon in terms of Application.
- CO2. To make aware the importance of sensor network and interoperability and Open Source Physical computing platform Arduino.
- CO3. To familiarize the programming languages such as Python and Raspberry Pi and extends its application to Software Defined Networking.
- CO4. To explore the relationship between IoT, Cloud Computing and Big Data.
- CO5. To understand the IoT in context of PLM

Bloom's Taxonomy: BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded

Summer-Term Examination (AY: 2024-25)

First Year M. Tech. (Semester-II)

Civil Engineering

Course: Open Elective: Operations Research

Course Code:

OEC-803

Date: 17-Jun-2025

Duration: 3 Hours

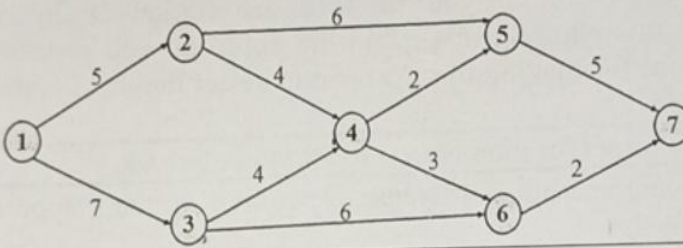
Time: 14:30 to 17:30

Max. Marks: 100

Instructions to Candidate:

1. Use of non-programmable calculator is allowed.
2. Question No. 01 is compulsory.
3. Solve any SIX (06) Questions from Question No. 02 to Question No. 09.

Q. No.	Questions	Marks	CO	BT
Question No. 01 is compulsory.				
Q. 1)	You want to decide a topic for your M. Tech. dissertation. Discuss the factors affecting this choice and explain the approach and procedure which you will use for making this decision to select the most suitable dissertation topic.	10	2, 5	3
Solve any SIX (06) Questions from Question No. 02 to Question No. 09.				
Q. 2)	a) Define systems engineering. Differentiate between systems approach and systems analysis.	07	1	1
	b) What is a model? State the different types of system models. Explain in brief the role of systems models.	08	3	2
Q. 3)	Solve the following LP problem using Simplex method: Maximize $Z = 4X_1 + 2.5X_2$ Subject to: $3X_1 + 5X_2 \leq 15$ $5X_1 + 2X_2 \leq 10$ $X_1 + X_2 \geq 1$; and $X_1 \geq 0, X_2 \geq 0$	15	2, 3	1
Q. 4)	a) How do real-life problems differ from the textbook type problems? Explain with an example.	07	1	1
	b) How do you justify the application of systems engineering to complex real-life problems? Discuss the role and significance of systems engineering.	08	2	2
Q. 5)	a) You have to buy a good laptop within a range of Rs. 50,000/- to 60,000/-. State the factors affecting, and discuss in details the problem solving procedure for making this decision to buy the best laptop.	10	1	1
	b) Explain the simplex algorithm using a neat sketch for solution to LP problems.	05	1	2
Q. 6)	a) Solve the following LP problem by graphical method (use graph paper on the last page of your answer sheet): Maximize $Z = 2X_1 + 3X_2$ Subject to: $-2X_1 + 4.5X_2 \leq 9$ $4.5X_1 - 3X_2 \leq 9$ $8X_1 + 6X_2 \geq 24$; and $X_1 \geq 0, X_2 \geq 0$ What shall be the change in solution if the nature of objective function is minimization instead of maximization?	10	3	3
	b) Explain the concept of system with a neat diagram.	05	4	2
Q. 7)	a) A total sum of Rs. 5 lakh is to be allocated amongst three business proposals. The option of not allocating any amount to any proposal is open and the money is to be distributed in units of Rs. 1 lakh. The returns from the three proposals from investment are given in the following	10	5	4

		table. Find the optimum allocation and the maximum returns using DP methodology.																															
		<table> <tr> <td>Investment →</td> <td>0</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> </tr> <tr> <td>Return from Proposal A</td> <td>0</td> <td>1</td> <td>2</td> <td>3</td> <td>3</td> <td>2</td> </tr> <tr> <td>Return from Proposal B</td> <td>0</td> <td>2</td> <td>3</td> <td>3</td> <td>4</td> <td>3</td> </tr> <tr> <td>Return from Proposal C</td> <td>0</td> <td>2</td> <td>2</td> <td>2</td> <td>4</td> <td>3</td> </tr> </table>	Investment →	0	1	2	3	4	5	Return from Proposal A	0	1	2	3	3	2	Return from Proposal B	0	2	3	3	4	3	Return from Proposal C	0	2	2	2	4	3			
Investment →	0	1	2	3	4	5																											
Return from Proposal A	0	1	2	3	3	2																											
Return from Proposal B	0	2	3	3	4	3																											
Return from Proposal C	0	2	2	2	4	3																											
	b)	State and explain the characteristics of a system.	05	2	1																												
Q. 8)	a)	A network is given in the figure. The shortest path between start node 1 and end node 7 is to be determined. Formulate a DP problem. Explicitly define the stages, states, and recursive relationship and then solve to find the optimal solution. 	10	3	3																												
	b)	What is ABC analysis? Explain its role in inventory system.	05	5	1																												
Q. 9)	a)	What do you understand by duality in LP? Explain its applications in brief.	08	3	1																												
	b)	Discuss the assumptions, limitations, and applicability of LP.	07	3	1																												

Course Outcomes (COs)

CO 1	Understand necessity and role of operation research in decision making for real-life problems.
CO 2	Analyze a real-life problem using system concepts and system models.
CO 3	Apply the concept and process of optimization.
CO 4	Formulate and solve problems using various operations research techniques.
CO 5	Practice operations research as a decision-making tool.

Bloom's Taxonomy (BT)

BT 1: Remember	BT 2: Understand	BT 3: Apply
BT 4: Analyze	BT 5: Evaluate	BT 6: Create